Name: _____

TOPIC TEST 2

Chapter 8: Arithmetic and geometric sequences

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 8 short answer questions (16 marks)
- Part C: 2 analysis questions (6 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 How many terms are in the sequence 23, 32, 41, …, 203?

A 4

B 21

C 23

D 41

E 203

[1 mark]

2 How many terms are in the sequence $-645, 215, -71\frac{2}{3}, \ldots, -\frac{215}{243}$?

A $\frac{215}{243}$

B 4

C 7

D 8

E -645

[1 mark]

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3 What is the first term that is less than zero in the series $301 + 289 + 277 + \cdots$?

A 0

B -1

C -11

D -12

E 26

[1 mark]

4 Find $S_8 - S_7$ for the geometric series $-\frac{1}{4} + \frac{1}{2} - 1 + \cdots$

A $-\frac{129}{12}$

B $\frac{126}{12}$

C $\frac{255}{12}$

D 16

E 32

[1 mark]

5 What is the limiting sum of $8 - 4 + 2 - 1 + \cdots$?

A $\frac{1}{2}$

B $5\frac{1}{3}$

C 8

D 15

E 16

[1 mark]

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Part B

8 short answer questions
16 marks
Show your working where appropriate.

1 Evaluate to 10 terms: $1 + 3 + 5 + 7 + \cdots$

[2 marks]

2 Evaluate to 6 terms: $19 + 21 + 23 + \cdots$

[2 marks]

3 For an AP with first term a, common difference d and n terms, write an expression for S_{n+1}.

[2 marks]

4 Insert an expression between the pair of variables p, q to form a geometric sequence.

[2 marks]

5 Find S_∞ for the geometric sequence $20, 8, \frac{16}{5}, \frac{32}{25}, \ldots$

[2 marks]

6 How many terms of the arithmetic sequence $2\frac{1}{2}, 4, 5\frac{1}{2}, \ldots$ must be taken for the sum to be greater than 200?

[2 marks]

7 Find the value of x such that $(3 - x)$, x, $(2 - x)$ are successive terms of a geometric sequence.

[2 marks]

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8 Write $0.6\overline{8}$ as a simplified fraction. Show all working.

[2 marks]

Part C

2 analysis questions
6 marks
Show your working where appropriate.

1 Jenny has invested $24 000 at 9% compounding annually. How much will her investment be worth after 10 years?

[2 marks]

2 Alan invested $100 each month for 20 years towards his superannuation.
His investment earned compound interest at a rate of 9% per annum paid monthly.

a How much did he invest over this time?

[1 mark]

b What was the value of his investment after 20 years?

[3 marks]
Total marks: 27

Answers

Part A

[1 mark each]

1 $t_n = a + (n-1)d$, $203 = 23 + (n-1) \times 9$, $n = 21$.

$\therefore$ **B** [1 mark]

2 $t_n = ar^{n-1}$, $-\frac{215}{243} = -645\ 3\left(-\frac{1}{3}\right)^{n-1}$, by trial and error $n = 7$.

$\therefore$ **C** [1 mark]

3 $t_n < 0$, $301 + (n-1) \times (-12) < 0$, $12n > 313$, $n > 26\frac{1}{12}$, $n = 27$. $\therefore t_{27} = -11$

$\therefore$ **C** [1 mark]

4 $S_8 - S_7 = t_8$, $t_8 = -\frac{1}{4} \times (-2)^{8-1} = 32$

$\therefore$ **E** [1 mark]

5 $S_\infty = \frac{a}{1-r}$, $S_\infty = \frac{8}{1-\left(-\frac{1}{2}\right)} = \frac{16}{2+1} = \frac{16}{3} = 5\frac{1}{3}$

$\therefore$ **B** [1 mark]

Part B

1 $1 + 3 + 5 + \cdots + 19$ [1 mark]

$S_{10} = \frac{19}{2}[1 + 19] = 190$ [1 mark]

2 $S = S_{15} - S_9$ [1 mark]

$= \frac{15}{2}[1 + 29] - \frac{9}{2}[1 + 17]$

$= 144$ [1 mark]

3 $S_n = \frac{n}{2}[2a + (n-1)d]$

$S_{n+1} = \frac{n+1}{2}[2a + (n+1-1)d]$ [1 mark]

$= \frac{n+1}{2}[2a + nd]$ or equivalent [1 mark]

4 Since p, x, q form part of a geometric sequence, a common ratio exists.

$\frac{q}{x} = \frac{x}{p}$ [1 mark]

$x^2 = pq$

$x = \pm\sqrt{pq}$ [1 mark]

5 $S_\infty = \frac{a}{1-r} = \frac{20}{1-\frac{2}{5}}$ [1 mark]

$= \frac{100}{5-2} = 33\frac{1}{3}$ [1 mark]

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6 $S_n > 200$

$\frac{n}{2}\left[5 + (n-1) \times \frac{3}{2}\right] > 200$ **[1 mark]**

$3n^2 + 4n - 800 > 0$

$n > 15.68, n < -17.01$

Since $n > 0$, $n = 16$ **[1 mark]**

7 $\frac{2-x}{x} = \frac{x}{3-x}$ **[1 mark]**

$x^2 = (2-x)(3-x)$

Solving for x, we get $x = 1\frac{1}{5}$ **[1 mark]**

8 $0.6\overline{8} = \frac{6}{10} + \frac{8}{100} + \frac{8}{1000} + \frac{8}{10\,000} + \cdots$ **[1 mark]**

$= \frac{6}{10} + \frac{\frac{8}{100}}{1 - \frac{1}{10}}$

$= \frac{6}{10} + \frac{8}{100 - 10} = \frac{6}{10} + \frac{8}{90}$

$= \frac{540 + 80}{900} = \frac{620}{900} = \frac{31}{45}$ **[1 mark]**

Part C

1 Investment $= (24\,000 \times 1.09) \times 1.09 \times 1.09 \times \cdots \times 1.09$ **[1 mark]**

$= 24\,000 \times 1.09^{10}$

$= \$56\,816.73$ **[1 mark]**

2 **a** Total outlay $= 100 \times 240$ **[1 mark]**

b Superannuation $= 100 \times 1.0075 + 100 \times 1.0075^2 + 100 \times 1.0075^3 + \cdots + 100 \times 1.0075^{240}$

$= 100 \times (1.0075 + 1.0075^2 + 1.0075^3 + \cdots + 1.0075^{240})$ **[1 mark]**

$= 100 \times \frac{1.0075\left(1.0075^{240} - 1\right)}{1.0075 - 1}$ **[1 mark]**

$= \$67\,289.60$ **[1 mark]**

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